Data Analytics & Visualisation

A Deep Dive into Data-Centric Engineering and Real-World Asset Telemetry

What is Data-Centric Engineering?

Physics-Informed

Unlike pure Data Science, DCE is constrained by the laws of physics and material science.

Structured Data

Transitioning from standalone reports to living, interconnected digital twin datasets.

Critical Decision-Making

Using high-fidelity data to manage risks in public and industrial infrastructure.

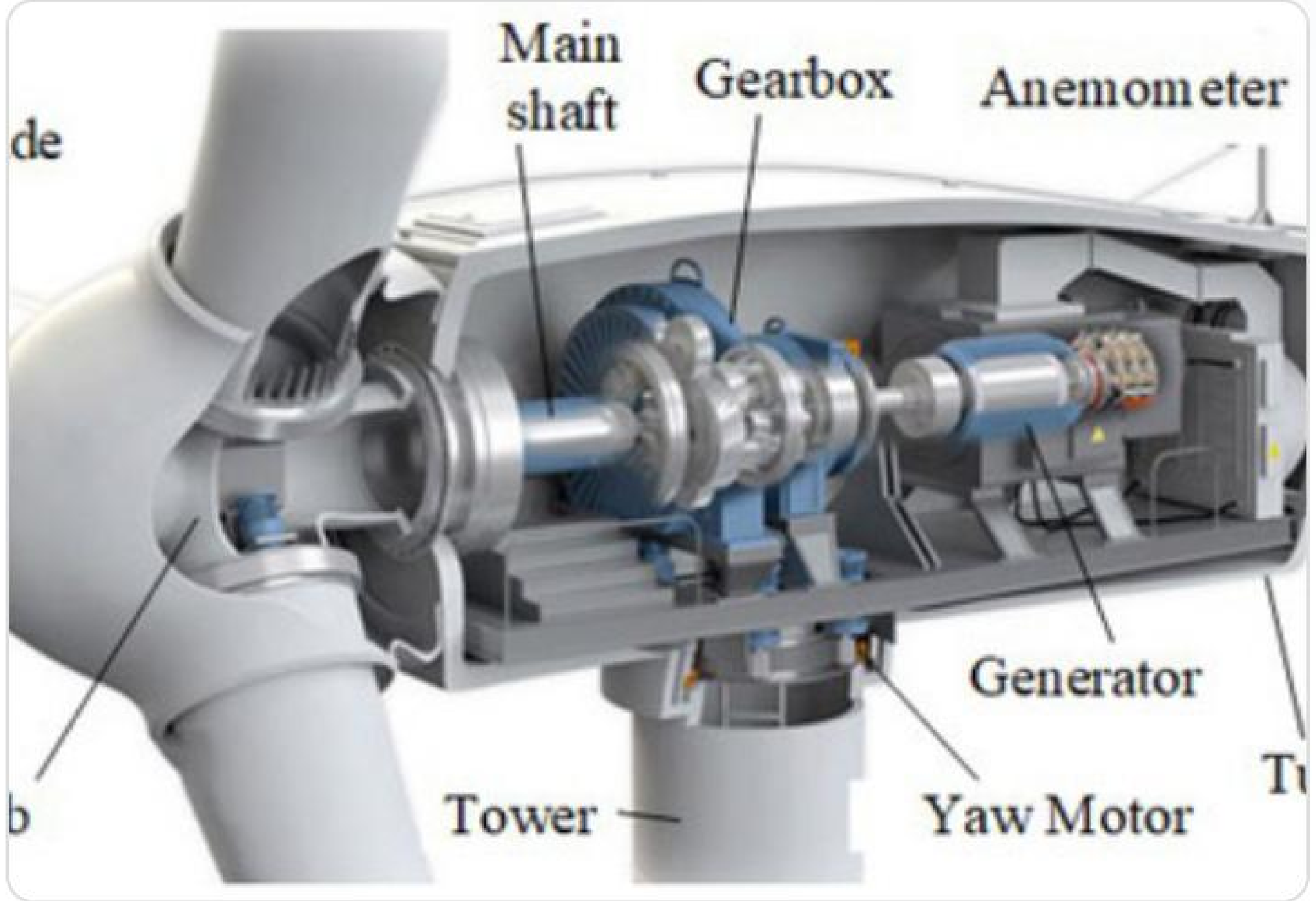
The Case Study: Wind Farm Fleet

WT-01: Healthy

Maintains stable, nominal physics across the 3-month timeline.

WT-02: Failed

Rapid degradation in Month 1 leading to catastrophic bearing seizure.



WT-03: Critical

Healthy initially, but begins tracking the failure signature of WT-02.

Section 1: The Ingestion Challenge

Messy Sensor Reality

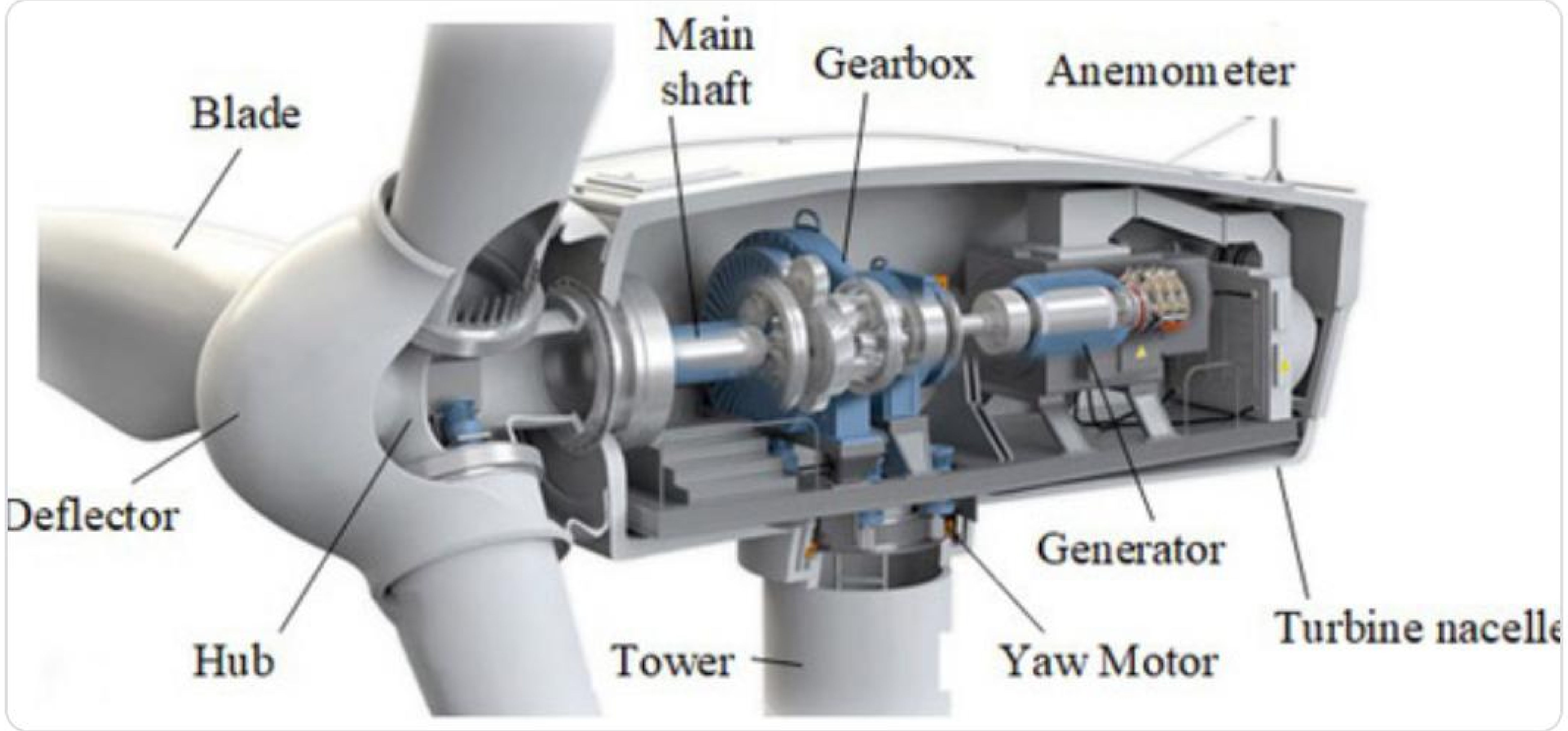
In Data-Centric Engineering, "bad" data is often the most valuable. Sensor dropouts and signal flatlines tell a story of physical failure.

We must ingest raw telemetry without losing the forensic evidence hidden in the noise.

Governance: Retention vs Omission

- **Forensic Review:** Keeping "failed" rows allows engineers to conduct post-mortem analysis on asset destruction.
- **Statistical Cleaning:** Creating separate datasets for clean statistical trends vs. raw anomaly logs.
- **Ethics of Omission:** Deleting sensor dropouts can mask safety risks and professional liabilities.
- **Data Lineage:** Tracking which sensors provided which readings and when they were last calibrated.

The Physical Triple Threat



RPM, Temperature, Vibration

These three variables form the "physics core" of our analysis:

RPM: The driver of energy production.
Temperature: Indicator of friction and lubrication breakdown.
Vibration: The fingerprint of structural degradation.

Section 2

Anomaly Hunting with Plotly

Catastrophe vs Early Signs

WT-02 Death Spiral

Observe the rapid linear climb in bearing temperature followed by a sudden drop in RPM. The interactive hover tools reveal the exact hour of the seizure.

WT-03 Physics Drift

While still spinning, WT-03 shows a gradual increase in Vibration relative to its load. This "drift" is invisible to static threshold alerts but clear in scatter plots.

Section 3: Normalisation

HEALTH RATIO

$$\frac{\text{Vibration}}{\text{RPM}}$$

Why Normalise?

Comparison is impossible when metrics use different scales (Celsius vs mm/s vs RPM). By calculating the **Vibration-to-RPM Ratio**, we create a speed-independent health index for our fleet.

The Operational Decision Matrix

Estimated Remaining Lifespan (Days)

Scenario: Budget allows only ONE overhaul this quarter. Where do we invest?

Ethics & Grid Priority

The $10k Choice

WT-02 is standard industrial revenue. WT-03 supplies a critical municipal **Hospital Grid**.

Does the risk of a blackout at a life-critical facility outweigh the lost revenue from a dead turbine? Dashboards are tools for accountability, not just performance.

Questions?

Thank you for participating in this DCE Workshop.

Contact: dce-workshop@breezetech.engineering

Image Sources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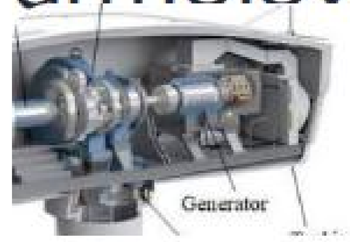

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